Config and add register Gitlab runner for group

1. Create rsa key format for gitlab-runner to registry and verify by nginx proxy
   mkdir -p /etc/gitlab-runner/client-certs
   cd /etc/gitlab-runner/client-certs

   # Create RSA key
   sudo openssl genrsa -traditional -out runner-client.key 2048
   sudo openssl req -new \ -key runner-client.key \ -subj "/CN=gitlab-runner/OU=DevSecOps/O=ZT" \ -out runner-client.csr

   # Create CSR
   sudo openssl req -new \ -key runner-client.key \ -subj "/CN=gitlab-runner/OU=DevSecOps/O=ZT" \ -out runner-client.csr
2. After create CSR
   # Copy CSR to Windows
   scp -P 2222 ubuntu@10.0.1.2:/etc/gitlab-runner/client-certs/runner-client.csr C:\Users\Administrator\Downloads\

   # Sign with template User (verify Client Authentication)
   certreq -submit -attrib "CertificateTemplate:User" `
   C:\Users\Administrator\Downloads\runner-client.csr `
   C:\Users\Administrator\Downloads\runner-client.crt

   # Copy cert to Linux server
   scp -P 2222 C:\Users\Administrator\Downloads\runner-client.crt ubuntu@10.0.1.2:/tmp/
3. Convert and verfity key format
   sudo mv /tmp/runner-client.crt /etc/gitlab-runner/client-certs/

   # Convert key sang RSA traditional format sudo openssl pkey \ -in /etc/gitlab-runner/client-certs/runner-client.key \ -out /etc/gitlab-runner/client-certs/runner-client-rsa.key \ -traditional

   # Verify MD5 match
   sudo openssl x509 -noout -modulus -in runner-client.crt | openssl md5 sudo openssl rsa -noout -modulus -in runner-client-rsa.key | openssl md5
4. Running Gitlab runner container

sudo docker run -d \ --name gitlab-runner \ --restart always \ --add-host "gitlab.zt.devsecops.local:10.0.1.2" \ -v /var/run/docker.sock:/var/run/docker.sock \ -v gitlab-runner-config:/etc/gitlab-runner \ gitlab/gitlab-runner:latest

5. Copy cert to container
   sudo docker exec -it gitlab-runner mkdir -p /etc/gitlab-runner/certs
   sudo docker cp /etc/ssl/certs/gitlab.crt \ gitlab-runner:/etc/gitlab-runner/certs/gitlab.zt.devsecops.local.crt
   # Client cert and key
   sudo docker cp /etc/gitlab-runner/client-certs/runner-client.crt \ gitlab-runner:/etc/gitlab-runner/runner-client.crt
   sudo docker cp /etc/gitlab-runner/client-certs/runner-client-rsa.key \ gitlab-runner:/etc/gitlab-runner/runner-client.key
6. Register Runner
   # Take token: Project → Settings → CI/CD → Runners → New project runner

   sudo docker exec -it gitlab-runner gitlab-runner register \ --url "https://gitlab.zt.devsecops.local" \ --token "TOKEN" \ --executor "docker" \ --docker-image "alpine:latest" \ --docker-volumes "/var/run/docker.sock:/var/run/docker.sock" \ --docker-network-mode "host" \ --tls-cert-file "/etc/gitlab-runner/runner-client.crt" \ --tls-key-file "/etc/gitlab-runner/runner-client.key" \ --non-interactive
7. Verify
   sudo docker exec -it gitlab-runner gitlab-runner list

```
ubuntu@active:/etc/gitlab-runner/client-certs$ sudo docker exec -it gitlab-runner gitlab-runner list
Runtime platform                                    arch=amd64 os=linux pid=78 revision=07e534ba version=18.9.0
Listing configured runners                          ConfigFile=/etc/gitlab-runner/config.toml
bd5afcd3e363                                        Executor=docker Token=glrt-_AeFpbvRlHwhbN3gR62rTm86MQpwOjIKdDozCnU6MQ8.01.1706y2g3e URL=https://gitlab.zt.devsecops.local
ubuntu@active:/etc/gitlab-runner/client-certs$
```